

# Simulation-based spatial optimisation of Solar PV for loss minimisation in smart grid distribution systems

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**Abstract:** The integration of distributed solar photovoltaic (PV) generation into radial distribution networks presents significant opportunities for power loss reduction, but only when panels are placed at network-optimal locations. Poorly sited PV units can induce reverse power flows, voltage violations, and paradoxically increased system losses. This paper presents a simulation-based spatial optimisation framework implemented in MATLAB to determine the optimal location and rated capacity of a single solar PV unit in a radial distribution network, minimising total active power losses. The IEEE 33-bus radial distribution system is adopted as the benchmark test case. A Backwards-Forward Sweep power flow solver based on the DistFlow formulation is implemented and validated against published benchmark results. A spatial sweep analysis evaluates the sensitivity of system losses to PV placement location at a fixed capacity of 1 MW. Particle Swarm Optimisation (PSO) is then applied to jointly optimise PV bus location and rated capacity across a continuous two-dimensional search space. Results demonstrate that the PSO-optimised placement of a 2.56 MW PV unit at Bus 6, the entry point of the most heavily loaded lateral branch, achieves a total active power loss of 102.84 kW, representing a 48.0% reduction from the baseline loss of 197.92 kW. Voltage profile analysis confirms that the baseline network violates the IEEE 0.95 pu lower limit at 14 buses, and that PSO-optimised placement resolves nearly all violations. A six-run sensitivity analysis confirms PSO robustness, yielding a standard deviation of only 0.94 kW across independent runs. These results establish that spatial placement is the dominant factor in PV-driven loss minimisation, with feeder branch entry points consistently outperforming deep-network locations across installed capacities.

**Key Terms:** Active power loss minimisation, DistFlow power flow, distributed generation, IEEE 33-bus system, loss reduction, MATLAB simulation, particle swarm optimisation, radial distribution network, solar photovoltaic placement, spatial optimisation, voltage profile, smart grid.

## 1. Introduction

Electricity consumption is rising much faster than overall energy use, with an estimated growth rate of around 3% [1]. This is roughly 2.3 times higher than the growth in total energy demand. Two sectors in particular, electric vehicles and data centres, showed especially rapid increases, with electricity use expanding by about 38% and 17% respectively. Even with this surge, traditional sectors such as industry, residential loads, and commercial buildings still account for most of global electricity consumption. In the United States, data centres alone were responsible for nearly half of the country's increase in electricity demand. For the first time, solar photovoltaic (PV) generation became the largest single contributor to global energy supply growth, accounting for more than one-quarter of the total increase in demand. Natural gas followed as the second-largest contributor at around 17%. Battery energy storage also experienced record expansion, adding approximately 110 GW of new capacity worldwide.

Additionally, more than 12 GW of new nuclear power construction began during the year. In terms of fossil fuels, global oil demand increased slightly (about 0.7%), though this was partially offset by reduced road-transport fuel consumption as electric vehicle adoption grew. Coal demand showed mixed trends: it declined in China as cleaner energy sources continued to scale up, but it increased in the United States, where high natural gas prices led some to switch back to coal.

Hence, solar energy emerges as the best option for future energy requirements. It is corroborated by several reasons, including its inexhaustibility and abundance. The Sun emits energy at the rate of  $3.8 \times 10^{23}$  kW, out of which approximately  $1.8 \times 10^{14}$  kW is intercepted by the Earth [2]. Further, its clean nature, improved voltage profile, and loss minimisation also contribute to better output efficiency than other energy sources. Solar PV's installed capacity is poised to surpass that of coal by 2027, making it the largest in the world. Cumulative solar PV capacity almost triples in our forecast, growing by almost 1500 GW over the period, exceeding natural gas by 2026 and coal by 2027. Annual solar PV capacity additions are expected to increase over the next five years [3].

Despite current higher investment costs due to elevated commodity prices, utility-scale solar PV is the least costly option for new electricity generation in a significant majority of countries worldwide. Distributed solar PV, such as rooftop solar on buildings, is also set for faster growth because of higher retail electricity prices and growing policy support to help consumers save money on their energy bills [3]

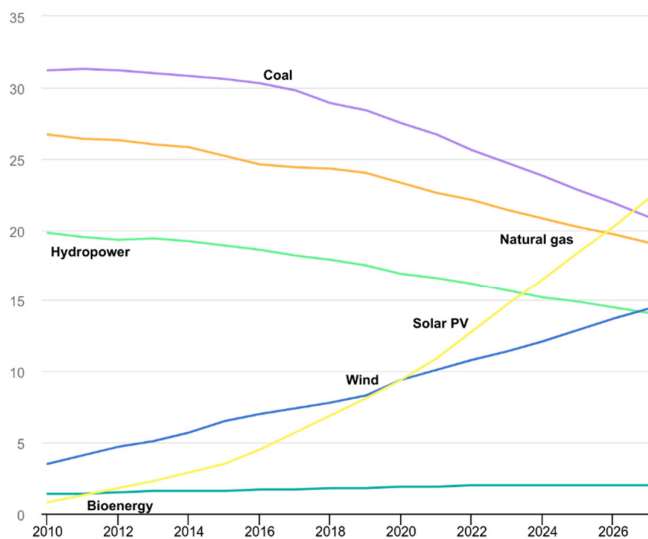


Figure 1: Share of cumulative power capacity by technology, 2010-2027 (Source: IEA)

To maximise output from solar PV panels, it is necessary to optimise the minimum number of panels required for a particular grid, while accounting for space, grid requirements, and grid stability. Incorrect siting and sizing of these PV modules can negatively affect the existing system's technical balance. This paper presents simulation results on the optimal placement and sizing of solar PV in a grid system, summarises comparative results, and outlines methods for optimising solar PV panels for loss minimisation.

## **2. Literature Review**

The optimal siting of solar photovoltaic (PV) generation within distribution networks has attracted substantial research interest, motivated by the measurable scale of distribution-level power losses- ranging from approximately 4% of generated energy in Germany to over 21% in India- and the well-documented capacity of strategically placed distributed generation (DG) to mitigate these losses [4]. The existing body of work spans three broadly distinguishable paradigms: metaheuristic optimisation applied to network bus selection, spatial and GIS-driven physical placement at the rooftop or terrain scale, and simulation frameworks that couple temporal load variability with PV output modelling. This review synthesises findings across these paradigms, with emphasis on methodology, quantitative outcomes, and the convergences and tensions that motivate the present study.

### **2.1 Metaheuristic Optimisation for Network-Level Placement**

The most extensively studied approach formulates PV placement as a constrained nonlinear optimisation problem in a radial distribution network, with minimisation of active and reactive power losses as the primary objective. The IEEE 33-bus and 69-bus radial distribution systems serve as near-universal benchmarks, and the forward-backwards sweep method is typically employed for load flow computation due to its numerical stability in radial topologies.

Particle Swarm Optimisation (PSO) is the most widely applied metaheuristic in this domain. On the IEEE 33-bus system, a PSO-based placement of a single 2440 kW PVDG at bus 9 achieved a 55.66% reduction in active power losses (from 205.9 kW to 91.7 kW) and a 52.78% reduction in reactive losses, alongside a 3.48% improvement in voltage profile at the weakest bus [5]. A constraint-handling variant of PSO applied to the same system yielded comparable reductions in active and reactive losses and produced annual cost savings exceeding those of competing algorithms, including the Teaching-Learning Artificial Bee Colony (TLABC) and the Pathfinder Algorithm, by 15.21% and 6.70%, respectively [6]. These studies confirm that PSO reliably converges to competitive solutions on standard radial test systems, though its performance is sensitive to parameter tuning and objective function weighting.

Genetic Algorithms (GAs) represent the principal alternative. When applied sequentially with increasing numbers of DGs on both the IEEE 33-bus and 69-bus systems, a GA-based approach using a Voltage Stability Index (VSI) to pre-screen candidate buses achieved loss reductions of 64.95% and 69.23%, respectively, with three distributed units — confirming that loss reduction scales monotonically with the number of optimally sited generators [7]. A direct PSO-vs-GA comparison on a modified IEEE 14-node network incorporating real hourly meteorological data from Tozeur, Tunisia, found that PSO converged faster and produced superior voltage and frequency profiles, though the margin narrowed under the dynamic 24-hour load profile condition [8]. This comparison is particularly instructive because it also demonstrated that single-snapshot optimisation, fixing load at a single operating point, produces misleading placement recommendations relative to temporally resolved optimisation, a methodological limitation that affects most earlier studies.

Hybrid algorithms have been proposed to capture the complementary strengths of different search paradigms. A Grey Wolf Optimiser-PSO (GWO-PSO) hybrid applied to the Dilla one feeder in Ethiopia, a practical 137-bus, 15.793 MW distribution network, outperformed standalone GWO, PSO, WOA, and SCA across a weighted multi-objective function combining active loss, reactive loss, and voltage deviation [9]. Three optimally sited PV-DG units provided the best trade-off between loss reduction and voltage improvement on this real feeder, and the study's use of an actual distribution network rather than an IEEE benchmark represents a methodologically significant departure from the norm. A comprehensive review of optimisation techniques for DG placement similarly concluded that hybrid metaheuristics consistently outperform single-algorithm approaches and identified multi-objective formulations combined with practical network constraints as the most productive current research direction [4].

## **2.2 Placement Incorporating PV Sizing and Storage**

Several studies have extended the placement problem to co-optimize PV sizing and, in some cases, the integration of battery energy storage systems (BESS). On the IEEE 33-bus system, a GA-based joint placement and sizing of a 715 kW PV unit at bus 9 reduced daily power losses to 2201.66 kW; the addition of a 90 kWh BESS at bus 16 brought losses to 2180.01 kW- a further reduction that, while modest in percentage terms, demonstrates the in-principle value of coordinated storage scheduling [10]. The study used a 24-hour German summer load profile, which is one of the few instances in the literature of temporally resolved optimisation applied to the joint placement-sizing-storage problem.

The importance of realistic PV modelling and network constraints on placement outcomes was demonstrated for a 69-bus IEEE test system using real meteorological data from Mashhad, Iran, across three load classes and three PV capacity configurations [11]. Incorporating reverse power flow constraints substantially reduced the feasible space of bus locations and PV sizes, and sun-tracking configurations consistently outperformed fixed-tilt across both loss-reduction and economic-return metrics. This finding highlights the degree to which placement recommendations are sensitive to modelling fidelity: simplified PV models without thermal and tracking dynamics can overestimate the benefit of installations at certain buses.

## **2.3 Centralised versus Decentralised Placement and Grid Impact**

An important dimension of placement strategy that is often elided in bus-optimisation studies is the centralisation-decentralisation trade-off. Across sunny summer, sunny winter, and cloudy summer scenarios, a centrally located PV matched to a commercial load profile whose demand coincides temporally with PV generation outperformed the decentralised residential configuration in voltage rise control, voltage fluctuation damping, and power loss reduction. Decentralised residential systems tended to be over-scaled relative to household peak consumption, generating surplus that increased feeder losses. This result underscores a finding that recurs across the literature: the temporal alignment between PV generation and local demand is as important a placement criterion as the electrical distance to load centres.

A complementary perspective is offered by community-scale residential optimisation using Mixed-Integer Linear Programming (MILP), which showed that deliberately distributing panels across east, west, and south roof facets rather than maximising south-facing area flattens net demand, reduces peak ramp rates, and decreases the battery storage capacity required to achieve grid balance [12]. While this study addresses the physical rooftop layer rather than the distribution network, it reinforces the principle that heuristic placement rules (e.g., orient south for maximum irradiance) are suboptimal when system-level objectives such as duck-curve mitigation or loss minimisation are considered.

## **2.4 Spatial and Simulation-Based Placement Frameworks**

A distinct body of work addresses the physical geometry of PV installation — where, on a rooftop or terrain, a panel is positioned, rather than which network bus it connects to. A GIS-based integer programming model extended the classical Maximal Covering Location Problem to the PV context, using high-resolution LiDAR data and ArcGIS Solar Analyst to identify viable rooftop cells and optimise panel siting and orientation simultaneously [13]. Flexible panel alignment across four orientation scenarios increased maximum energy production by up to 6% relative to strict alignment requirements, with the conflict-zone formulation effectively enforcing non-overlap constraints at a realistic rooftop scale.

An AI-driven simulation framework integrating NASA POWER irradiance data, OpenStreetMap geometry, and Digital Elevation Model (DEM) terrain data applied a Genetic Algorithm to dynamic tilt optimisation and reported 20–25% energy gains over fixed-tilt configurations [14]. The study demonstrated the framework's global applicability but employed a simple linear-regression feasibility classifier and did not validate energy predictions against measured plant output, a limitation that illustrates the gap between feasibility screening and performance quantification in data-driven approaches. A broad survey of optimal placement methods for renewable energy sources identified PSO and Biogeography-Based Optimisation (BBO) as the most common metaheuristics, the Jensen wake model as the most-used wake model for wind, and noted that solar-specific placement research remains comparatively underdeveloped relative to wind, with no unified benchmark or ontology in common use [15].

## **2.5 Synthesis and Research Gaps**

Taken together, the literature reveals a field that has converged methodologically: PSO and its hybrids applied to radial IEEE test networks with backwards-forward sweep load flow, but remains fragmented in important dimensions. Almost universally, distribution network placement studies assume deterministic or single-snapshot PV output, neglecting the stochasticity of irradiance and temperature that shifts optimal bus location and size under real atmospheric variability. Only a small subset of studies uses hourly or sub-hourly temporal resolution, and even fewer jointly optimise placement, sizing, storage scheduling, and reactive power dispatch. The near-exclusive reliance on IEEE benchmark networks means that placement strategies are rarely validated against the topological and load characteristics of actual smart grid distribution feeders.

Equally, the physical and network layers of the placement problem remain largely decoupled: studies that optimise rooftop geometry and terrain shading do not connect their outputs to bus-level network optimisation, and vice versa. The present work addresses this gap by embedding a simulation-based spatial optimisation framework directly within a smart grid distribution network model, enabling the evaluation of loss minimisation objectives under realistic load profiles and PV generation variability; thereby bridging the spatial placement and network optimisation paradigms that the existing literature treats in isolation.

### 3. Methodology

This section presents the complete simulation framework developed to evaluate the spatial impact of solar PV placement on active power losses in a radial distribution network. The framework comprises five sequential stages: network data specification, power flow analysis, PV modelling, spatial sweep evaluation, and metaheuristic optimisation using Particle Swarm Optimisation (PSO). All stages were implemented in MATLAB R2020a.

#### 3.1 Test System: IEEE 33-Bus Distribution Network

The IEEE 33-bus radial distribution system serves as the benchmark test case for this study. The network operates at a base voltage of 12.66 kV and comprises 33 buses connected by 32 branches in a radial (tree) topology. The total connected active load is 3.715 MW with a reactive load of 2.300 MVAR. The network consists of one main feeder running from the substation (Bus 1) through to Bus 18, with three lateral branches: Lateral 1 branching from Bus 2 to serve buses 19–22, Lateral 2 branching from Bus 3 to serve buses 23–25, and Lateral 3 branching from Bus 6 to serve buses 26–33. Lateral 3 represents the longest and most heavily loaded branch in the system, carrying a disproportionate share of total reactive demand.

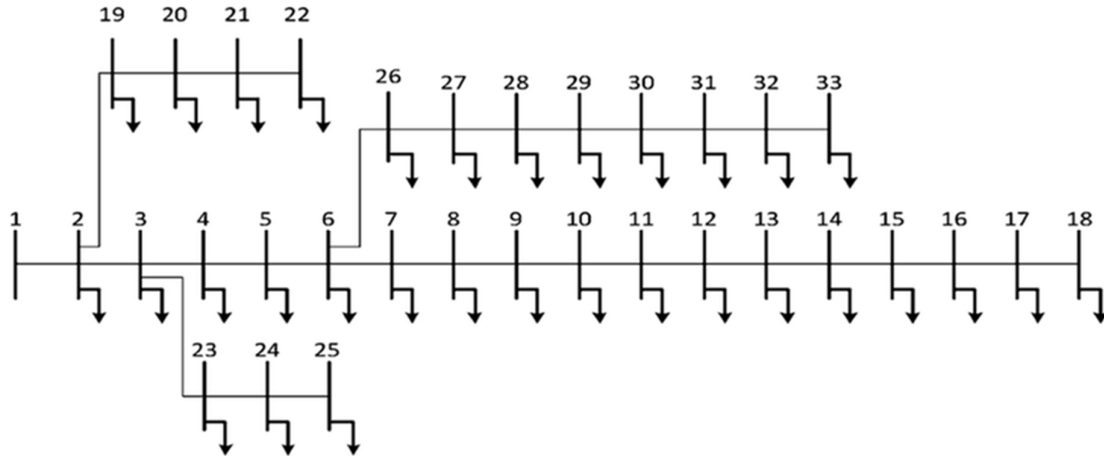


Figure 2: Diagram of IEEE 33-Bus Radial Distribution System

Bus 1 is designated as the slack bus, with voltage fixed at 1.0 per unit (pu) representing the substation maintaining nominal voltage. All remaining buses are load buses whose voltages are determined by the power flow solution. The system parameters- branch resistance, reactance, and nodal active and reactive loads- are taken directly from the original published data and coded as input matrices in MATLAB.

| Parameter                     | Value                                  |
|-------------------------------|----------------------------------------|
| Number of Buses               | 33                                     |
| Number of Branches            | 32                                     |
| Base Voltage                  | 12.66 kV                               |
| Total Active Load             | 3.715 MW                               |
| Total Reactive Load           | 2.300 MVAR                             |
| Baseline Active Power Loss    | 202 kW (published); 197.92 (simulated) |
| Minimum Baseline Voltage      | 0.913 pu at Bus 18                     |
| Buses Violating 0.95 pu limit | 14                                     |

Table 1: IEEE 33-Bus System Summary Parameters

### 3.2 Power Flow Analysis: DistFlow Backwards-Forward Sweep

Power flow analysis is the computational core of the simulation framework, used to determine bus voltages and total system active power losses under a given loading and generation condition. For radial distribution networks, the Backwards-Forward Sweep (BFS) method based on the DistFlow branch power-flow formulation is preferred over the Newton-Raphson method. This choice is motivated by the radial topology of the test network, for which BFS offers superior numerical stability, faster convergence, and lower computational overhead compared to full Jacobian-based methods.

The DistFlow formulation expresses network equations in terms of branch active and reactive power flows and squared voltage magnitudes, thereby avoiding complex-number arithmetic. For a branch connecting sending bus  $i$  and receiving bus  $j$  with resistance  $R$  and reactance  $X$ , the forward sweep voltage update equation is given by:

$$V^2(j) = V^2(i) - 2(R \cdot P_{ij} + X \cdot Q_{ij}) + (R^2 + X^2)(P_{ij}^2 + Q_{ij}^2)/V^2(i) \quad (1)$$

where  $P_{ij}$  and  $Q_{ij}$  denote the active and reactive power flowing from bus  $i$  to bus  $j$ , respectively. In the backward sweep, branch power flows are accumulated from the terminal buses toward the source, incorporating downstream loads and losses. In the forward sweep, bus voltages are updated from the source outward using equation (1). Iteration continues until the maximum bus voltage change between successive iterations falls below a convergence tolerance of  $1 \times 10^{-6}$  pu. Total active power loss is computed from the converged solution as:

$$P_{loss} = \sum_k R_k \cdot (P_k^2 + Q_k^2)/V^2(\text{sending bus}) \quad (2)$$

where the summation is taken over all 32 network branches  $k$ . The power flow solver was implemented in MATLAB and validated against the published IEEE 33-bus benchmark results of Baran and Wu [1]. The simulated baseline total active power loss of 197.92 kW compares favourably with the published value of approximately 202 kW, resulting in a discrepancy of less than 2% and within the acceptable tolerance for benchmark validation.

### 3.3 Solar PV Modelling

Solar photovoltaic generation is modelled as a controllable active power injection at a candidate bus. In the power flow framework, this is implemented by treating the PV unit as a negative active load at the specified bus, reducing the net active power demand at that node:

$$P_{net}(bus) = P_{load}(bus) - P_{PV} \quad (3)$$

where  $P_{PV}$  denotes the rated active power output of the PV unit in kilowatts. Unity power factor is assumed throughout this study, meaning the PV inverter injects only active power, with reactive power injection set to zero. This represents a standard, commercially available solar inverter that operates without voltage-variance (volt-VAR) control capability. The implications of this assumption are discussed in Section V.

No time-varying solar irradiance profile is modelled; the PV unit is treated as a dispatchable source operating at its rated output. This is consistent with the adopted static load flow formulation and represents worst-case or peak-generation conditions, which are the most relevant scenarios for loss minimisation studies.

### 3.4 Spatial Sweep Analysis

To characterise the sensitivity of total system losses to PV placement location, independent of capacity effects, a spatial sweep was conducted by placing a fixed 1 MW PV unit at each candidate bus from Bus 2 to Bus 33. For each placement, the power flow was re-executed and the resulting total active power loss recorded. Bus 1 was excluded because it represents the slack bus (substation), where PV injection is physically meaningless. This produced a loss-versus-location profile across all 32 candidate buses, enabling direct identification of the network locations where distributed generation is most effective at reducing resistive losses.

The spatial sweep provides both a practical result, identifying the best single-unit placement for a fixed capacity, and a physically interpretable map of network sensitivity that informs the subsequent optimisation stage.

### 3.5 Particle Swarm Optimisation

While the spatial sweep identifies the optimal location for a fixed PV capacity, it does not address the joint optimisation of both placement location and rated capacity. To simultaneously determine the optimal bus location and PV size, Particle Swarm Optimisation (PSO) [16] was implemented. PSO is a population-based metaheuristic algorithm inspired by the collective foraging behaviour of bird flocks, in which candidate solutions (particles) navigate a multi-dimensional search space guided by their own historical best performance and the best performance observed across the entire swarm.

Each particle represents a candidate solution defined by a two-dimensional position vector  $[bus, P_{PV}]$ , where  $bus \in \{2, 3, \dots, 33\}$  and  $P_{PV} \in [100, 3000]$  kW. The fitness function evaluated for each particle is the total active power loss returned by the power flow solver for the corresponding PV placement and capacity. The particle velocity update equation governing swarm dynamics is:

$$v(t+1) = w \cdot v(t) + c^1 \cdot r^1 \cdot (x_p best - x(t)) + c^2 \cdot r^2 \cdot (x_g best - x(t)) \quad (4)$$

where  $v(t)$  and  $x(t)$  are the velocity and position vectors at iteration  $t$ ,  $x_p best$  is the particle's personal best position,  $x_g best$  is the global best position found by any particle,  $r_1$  and  $r_2$  are independent uniform random numbers in  $[0,1]$ ,  $w$  is the inertia weight, and  $c_1$  and  $c_2$  are the



cognitive and social acceleration coefficients, respectively. Position is updated as  $x(t+1) = x(t) + v(t+1)$ , with bus position rounded to the nearest integer and both dimensions clamped to their respective bounds after each update.

The PSO hyperparameters were set as follows based on established guidelines [17]: swarm size  $N = 30$  particles, maximum iterations  $T = 100$ , inertia weight  $w = 0.7$ , cognitive coefficient  $c_1 = 1.5$ , and social coefficient  $c_2 = 1.5$ . The algorithm was implemented from first principles in MATLAB without the use of any optimisation toolboxes. To assess solution robustness given PSO's stochastic nature, the algorithm was executed 6 times independently with fresh random initialisation, and the variance in optimal solutions across runs was recorded.

| Parameter                  | Value                        |
|----------------------------|------------------------------|
| Swarm Size (N)             | 30 Particles                 |
| Maximum Iterations (T)     | 100                          |
| Inertia Weight (w)         | 0.7                          |
| Cognitive Coefficient (c2) | 1.5                          |
| Bus Search Space           | 2-33 (integer)               |
| PV Capacity Search Space   | 100-3000 kW (continuous)     |
| Fitness Function           | Total Active Power Loss (kW) |
| Independent Runs           | 6                            |

Table 2: PSO Algorithm Parameters

## 4. Experiments and Results

This section presents the results of the three experimental stages: power flow baseline validation, spatial sweep analysis, and PSO optimisation, followed by a comparative analysis of voltage profiles and a sensitivity evaluation.

### 4.1 Baseline Power Flow Validation

The DistFlow power flow solver was first executed on the unmodified IEEE 33-bus system with no PV injection to establish the baseline operating condition. The simulation produced a total active power loss of 197.92 kW, which corresponds to a loss rate of approximately 5.3% relative to the 3.715 MW total load. This result compares closely with the published benchmark value of approximately 202 kW [18], with a discrepancy of less than 2%, confirming the validity of the implemented power flow formulation.

Voltage analysis of the baseline solution revealed severe undervoltage conditions across a substantial portion of the network. The minimum bus voltage was 0.913 pu at Bus 18, at the terminal end of the main feeder, representing a 8.7% deviation below nominal. A total of 14 buses exhibited voltages below the IEEE standard lower limit of 0.95 pu. These violations were concentrated in two regions: the distal segment of the main feeder (buses 13–18) and the full extent of Lateral 3 (buses 28–33), both of which are electrically remote from the substation and subject to the greatest cumulative voltage drop.

## 4.2 Spatial Sweep Results

The spatial sweep was conducted by placing a 1 MW PV unit at each of the 32 candidate buses (buses 2–33) and recording the resulting total active power loss. The loss-versus-location profile exhibits a physically interpretable pattern consistent with the network topology.

Placement near the substation (buses 2–4) provided only a minimal reduction in losses, with losses remaining above 165 kW. This is expected, as power injection close to the source reduces line currents only over short segments of the feeder. Moving deeper into the network, particularly into Lateral 3, progressively reduced losses, reflecting the greater cumulative resistance over which the current reduction is achieved. The minimum loss under the spatial sweep was achieved at Bus 30, located in the middle segment of Lateral 3, with a total loss of 125.35 kW, representing a 36.67% reduction from the baseline of 197.92 kW.

Notably, placement at buses on Lateral 1 (buses 19–22) produced losses approaching or exceeding the baseline, with Bus 22 recording 203.58 kW—higher than the no-PV case. This phenomenon arises from reverse power flow: at unity power factor, injecting 1 MW into a bus serving a low local load results in excess generation flowing back toward the substation, increasing the current magnitude in upstream lines and thereby increasing resistive losses. This result demonstrates that PV placement without spatial optimisation can actively degrade network performance.

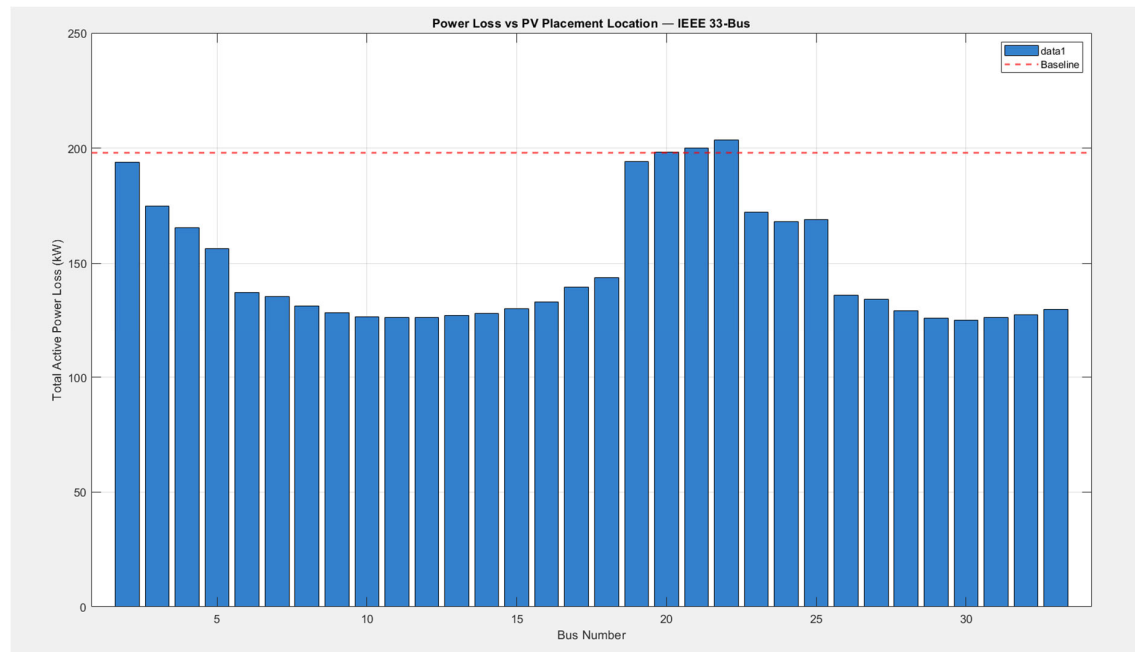


Figure 3: Total active power loss as a function of PV placement location for a fixed 1 MW injection (IEEE 33-bus system). The dashed line indicates the 197.92 kW baseline.

## 4.3 PSO Optimisation Results

PSO was applied to jointly optimise PV bus location and rated capacity. Across six independent runs, the algorithm consistently converged to one of two near-optimal solutions: Bus 6 at

approximately 2562 kW, and Bus 26 at approximately 2423 kW. These two solutions represent the global and local optima of the loss landscape, respectively, corresponding to the entry point and mid-section of Lateral 3.

The global optimum- Bus 6 at 2561.54 kW, achieved a total active power loss of 102.84 kW, representing a 48.04% reduction from the baseline of 197.92 kW. This solution was identified in four of six independent runs. The alternative solution- Bus 26 at 2422.76 kW, achieved a 104.67 kW (47.12%) reduction in loss and was identified in the remaining two runs. The existence of two competing optima is physically consistent: both buses are located at or near the entry point of Lateral 3, and both benefit from the same mechanism of supplying the heavy downstream load locally. Bus 6, as the root of Lateral 3, provides a marginally superior solution because PV injection there simultaneously reduces current in the feeder segment from Bus 2 to Bus 6, as well as in all downstream branches.

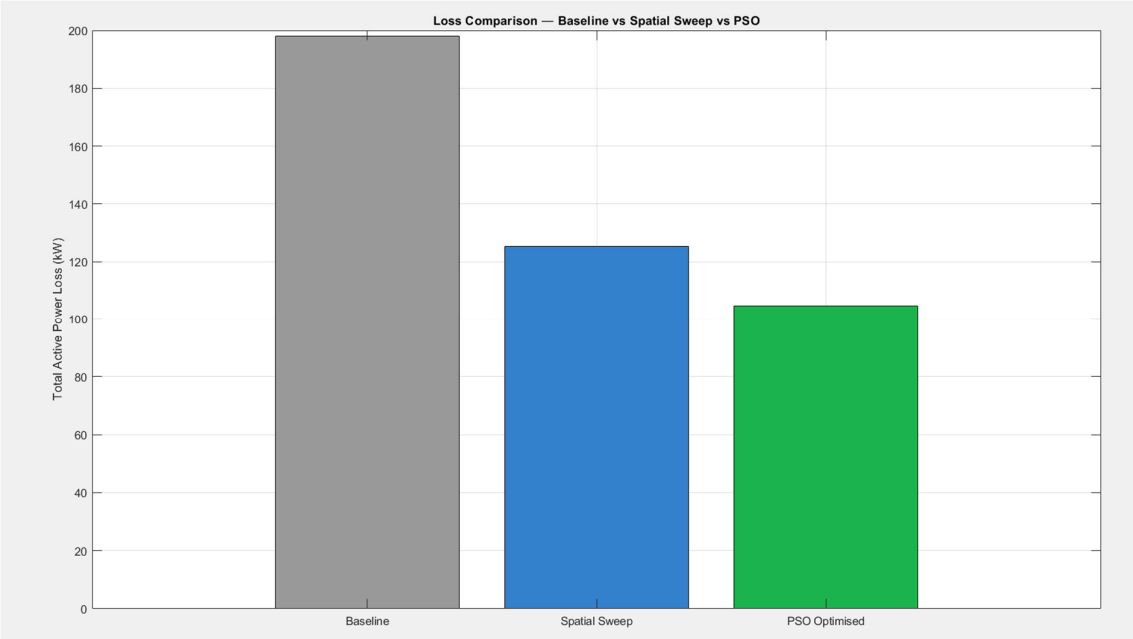


Figure 4: Comparative total active power loss for three scenarios: baseline (no PV), spatial sweep optimal (Bus 30, 1 MW), and PSO optimised (Bus 6, 2.56 MW).

| Run | Optimal Bus | PV Size (kW) | Loss (kW) | Reduction (%) |
|-----|-------------|--------------|-----------|---------------|
| 1   | Bus 26      | 2422.76      | 104.67    | 47.12         |
| 2   | Bus 6       | 2561.54      | 102.84    | 48.04         |
| 3   | Bus 6       | 2561.54      | 102.84    | 48.04         |
| 4   | Bus 26      | 2422.76      | 104.67    | 47.12         |
| 5   | Bus 6       | 2561.54      | 102.84    | 48.04         |
| 6   | Bus 6       | 2561.54      | 102.84    | 48.04         |

Table 3: PSO Sensitivity Analysis- Six Independent Runs

Mean Loss: 103.45 kW | Std Dev: 0.94 kW | Best: 102.84 kW Mean Reduction: 47.73%

#### 4.4 Voltage Profile Analysis

Fig. 5 presents the per-unit voltage magnitudes at each bus for three scenarios: baseline (no PV), spatial-sweep optimal (Bus 30, 1 MW), and PSO-optimised (Bus 6, 2.56 MW). IEEE standard voltage limits of 0.95 pu and 1.05 pu are overlaid for reference.

The baseline voltage profile exhibits a monotonically decreasing trend from Bus 1 (1.0 pu) along the main feeder, reaching a minimum of 0.913 pu at Bus 18. Fourteen buses fall below the 0.95 pu lower limit, concentrated in the distal main feeder and the full length of Lateral 3. The spatial sweep solution (Bus 30, 1 MW) improves voltages along Lateral 3 but provides limited benefit to the main feeder, and lateral violations largely persist. The PSO-optimised solution produces a markedly superior voltage profile. By injecting 2.56 MW at Bus 6, the root of Lateral 3, the solution simultaneously reduces current draw from the substation across the entire main feeder and provides local generation to support all downstream buses in Lateral 3. The resulting voltage profile remains above 0.95 pu at nearly all buses, with only marginal violations at the distal ends of laterals. The minimum bus voltage improves from 0.913 pu (baseline) to approximately 0.956 pu under PSO optimisation.

A notable feature of the PSO voltage profile is a visible local elevation at Bus 19. Bus 19 is the entry point of Lateral 1, which branches directly from Bus 2. When the PV injection at Bus 6 elevates voltage along the main feeder, Bus 2—already close to the substation—experiences an above-nominal voltage. Bus 19, inheriting its voltage directly from Bus 2, exhibits a corresponding local spike. This is a boundary effect arising from the interaction between the fixed slack bus voltage at Bus 1 and upstream propagation of the PV injection, and is consistent with expected network physics.

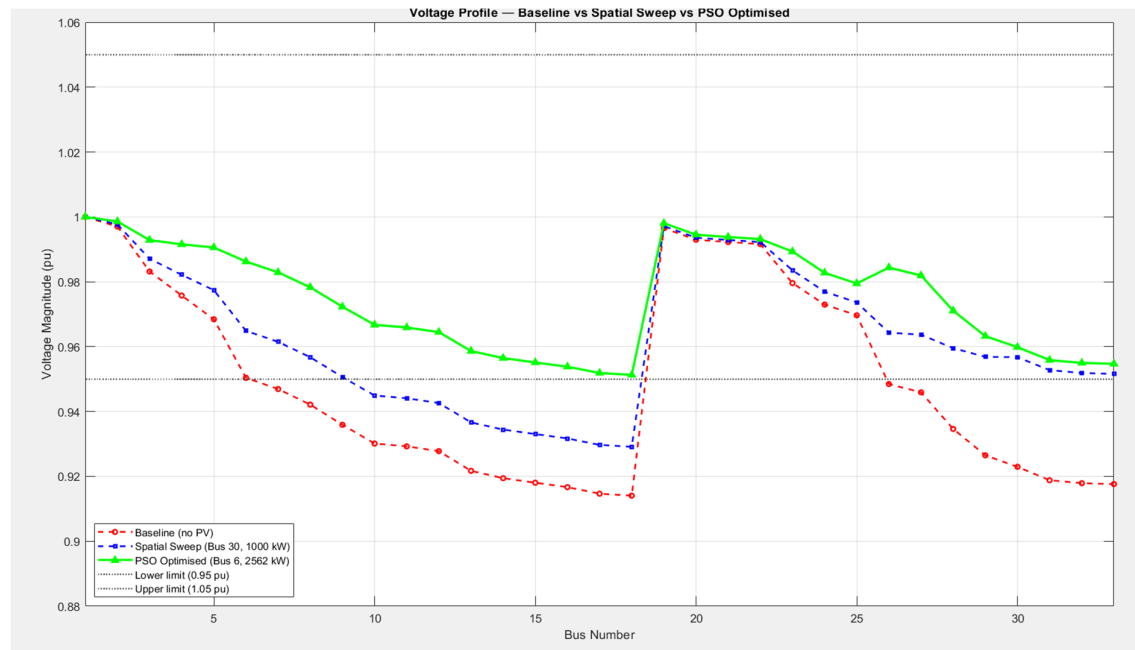


Figure 5: Per-unit voltage magnitude profile across all 33 buses for three scenarios: baseline, spatial sweep optimal (Bus 30, 1 MW), and PSO optimised (Bus 6, 2.56 MW). IEEE limits of 0.95 pu and 1.05 pu are shown as dotted reference lines.

| Scenario         | PV Location | PV Size (kW) | Total Loss (kW) | Reduction (%) |
|------------------|-------------|--------------|-----------------|---------------|
| Baseline (no PV) | -           | -            | 197.92          | -             |
| Spatial Sweep    | Bus 30      | 1000         | 125.35          | 36.67         |
| PSO Optimised    | Bus 6       | 2561.54      | 102.84          | 48.04         |

Table 4: Summary of Results Across All Scenarios

## 5. Discussion

The results establish several important findings regarding distributed PV integration in radial distribution networks. First and most significantly, spatial placement is the dominant determinant of loss minimisation outcomes. The comparison between the spatial sweep (Bus 30, fixed 1 MW, 36.67% reduction) and PSO (Bus 6, 2.56 MW, 48.04% reduction) demonstrates that a well-located PV unit of any reasonable size consistently outperforms a poorly located unit even at substantially higher capacity. This finding has direct practical implications: network operators and planners should prioritise optimal siting before scaling capacity.

Second, feeder branch entry points are consistently identified as superior PV locations. Both PSO optima, Bus 6 and Bus 26, are located at or near the root of Lateral 3, the network's most heavily loaded branch. This pattern is physically intuitive- injection at a branch entry point simultaneously reduces current in all downstream segments and in the upstream main feeder segments feeding that branch, whereas injection deeper within a branch benefits only the segments beyond the injection point. The implication for practical planning is that PV siting studies should prioritise the evaluation of feeder-branch interconnection points.

Third, the unity-power-factor assumption adopted in this study is a deliberate simplification. Modern smart inverters can operate with leading or lagging power factor, enabling simultaneous reactive power injection (volt-VAR control). Extending the PSO optimisation to a three-dimensional search space, jointly optimising bus location, active power, and reactive power injection, would likely yield further loss reduction and more complete resolution of voltage violations. This represents the most directly impactful extension of the present work.

Fourth, the static load model employed here does not capture the temporal variability of both network loading and solar irradiance. In a real network, PV output varies continuously with solar conditions, while load profiles change throughout the day and across seasons. Dynamic optimisation approaches, including time-series simulation and multi-period OPF formulations, would be necessary to fully characterise the operational impact of spatially optimised PV across varying conditions.

## 6. Conclusion

This paper has presented a simulation-based spatial optimisation framework for determining the optimal location and rated capacity of a solar PV unit in a radial distribution network to minimise total active power losses. Using the IEEE 33-bus system as a benchmark, a DistFlow Backwards-Forward Sweep power flow solver was implemented and validated in MATLAB, followed by a systematic spatial sweep analysis and Particle Swarm Optimisation of joint location and capacity variables.

The key findings are fourfold. First, PSO-optimised placement of a 2.56 MW PV unit at Bus 6 achieves a 48.0% reduction in total active power losses compared to the no-PV baseline, improving from 197.92 kW to 102.84 kW. Second, voltage profile analysis demonstrates that the baseline network experiences violations at 14 buses below the IEEE 0.95 pu lower limit, and that PSO-optimised placement resolves nearly all violations, with minimum bus voltage improving from 0.913 pu to approximately 0.956 pu. Third, spatial placement is more consequential than installed capacity: feeder branch entry points, specifically Bus 6 and Bus 26, both at the root of the most heavily loaded lateral, consistently outperform deep-network locations regardless of capacity. Fourth, PSO is demonstrated to be a robust and reliable optimisation tool for this class of problem, with six independent runs yielding a standard deviation of only 0.94 kW, confirming consistent convergence to near-optimal solutions despite stochastic initialisation.

The framework developed in this work is directly extensible to larger networks, multi-unit placement problems, and joint active-reactive power optimisation. Future work will incorporate reactive power injection modelling via smart inverter volt-VAR control, time-varying load and irradiance profiles, battery energy storage co-optimisation, and multi-unit PV siting formulations. The results of this study provide a quantitative, simulation-validated foundation for evidence-based distributed solar PV planning in smart grid distribution systems.

## 7. References

- [1] “Global Energy Review 2025.” [Online]. Available: [www.iea.org](http://www.iea.org)
- [2] N. Kannan and D. Vakeesan, “Solar energy for future world: - A review,” Sep. 01, 2016, *Elsevier Ltd.* doi: 10.1016/j.rser.2016.05.022.
- [3] I. - International Energy Agency, “Renewables 2022,” 2022. [Online]. Available: [www.iea.org](http://www.iea.org)
- [4] S. A. Adegoke, Y. Sun, A. S. Adegoke, and D. Ojeniyi, “Optimal placement of distributed generation to minimize power loss and improve voltage stability,” Nov. 15, 2024, *Elsevier Ltd.* doi: 10.1016/j.heliyon.2024.e39298.
- [5] R. Gnanendar and M. Sushama, “International Journal of INTELLIGENT SYSTEMS AND APPLICATIONS IN ENGINEERING Enhancing Distribution Network Efficiency and Symmetry via Optimal Sizing and Location of Photovoltaic DG Using PSO.” [Online]. Available: [www.ijisae.org](http://www.ijisae.org)

- [6] M. A. Ali, A. R. Bhatti, A. Rasool, M. Farhan, and E. Esenogho, "Optimal Location and Sizing of Photovoltaic-Based Distributed Generations to Improve the Efficiency and Symmetry of a Distribution Network by Handling Random Constraints of Particle Swarm Optimization Algorithm," *Symmetry (Basel)*, vol. 15, no. 9, Sep. 2023, doi: 10.3390/sym15091752.
- [7] S. Prakash, V. Rangta, N. S. Jayalakshmi, and V. K. Jadoun, "Optimal Location of DGs in a Distribution System for Maintaining Voltage Profile and Loss Reduction Using Genetic Algorithm," in *2020 International Conference on Power Electronics and IoT Applications in Renewable Energy and its Control, PARC 2020*, Institute of Electrical and Electronics Engineers Inc., Feb. 2020, pp. 59–64. doi: 10.1109/PARC49193.2020.236558.
- [8] I. Khenissi, R. Sellami, M. A. Fakhfakh, and R. Neji, "Power Loss Minimization Using Optimal Placement and Sizing of Photovoltaic Distributed Generation Under Daily Load Consumption Profile with PSO and GA Algorithms," *Journal of Control, Automation and Electrical Systems*, vol. 32, no. 5, pp. 1317–1331, Oct. 2021, doi: 10.1007/s40313-021-00744-7.
- [9] A. B. Alyu, A. O. Salau, B. Khan, and J. N. Eneh, "Hybrid GWO-PSO based optimal placement and sizing of multiple PV-DG units for power loss reduction and voltage profile improvement," *Sci. Rep.*, vol. 13, no. 1, Dec. 2023, doi: 10.1038/s41598-023-34057-3.
- [10] M. Dewantara, L. M. Putranto, R. Irnawan, and Sarjiya, "Minimization of Power Losses through Optimal Placement and Sizing from Solar Power and Battery Energy Storage System in Distribution System," in *2020 3rd International Seminar on Research of Information Technology and Intelligent Systems, ISRITI 2020*, Institute of Electrical and Electronics Engineers Inc., Dec. 2020, pp. 400–405. doi: 10.1109/ISRITI51436.2020.9315446.
- [11] S. Ahmadnia, E. Tafahi, and F. Shakhshi Dastgahian, "Optimal Placement and Sizing for Solar Farm with Economic Evaluation, Power Line Loss and Energy Consumption Reduction," *IETE J. Res.*, vol. 68, no. 3, pp. 2175–2190, 2022, doi: 10.1080/03772063.2019.1694450.
- [12] A. Pigott, K. Baertlein, and K. Baker, "Optimizing Solar Panel Placement through Passive Architectural Design," *Journal of Architectural Engineering*, vol. 29, no. 4, Dec. 2023, doi: 10.1061/jaeied.aeeng-1649.
- [13] Q. Zhong and D. Tong, "Spatial layout optimization for solar photovoltaic (PV) panel installation," *Renew. Energy*, vol. 150, pp. 1–11, May 2020, doi: 10.1016/j.renene.2019.12.099.
- [14] D. Bhatt, P. K. Sharma, and C. Parmar, "AI-Driven Solar Panel Placement Optimization," in *2025 Seventh International Conference on Research in Computational*

*Intelligence and Communication Networks (ICRCICN)*, IEEE, Dec. 2025, pp. 258–262.  
doi: 10.1109/ICRCICN68210.2025.11364963.

- [15] M. Y. Khan, M. Ali, S. Qaisar, M. Naeem, C. Chrysostomou, and M. Iqbal, “Placement Optimization for Renewable Energy Sources: Ontology, Tools, and Wake Models,” *IEEE Access*, vol. 8, pp. 72781–72800, 2020, doi: 10.1109/ACCESS.2020.2984901.
- [16] J. Kennedy, R. Eberhart, and bls gov, “Particle Swarm Optimization.”
- [17] Y. Shi and R. Eberhart, “A Modified Particle Swarm Optimizer.”
- [18] E. Baran and F. F. Wu, “NETWORK RECONFIGURATION IN DISTRIBUTION SYSTEMS FOR LOSS REDUCTION AND LOAD BALANCING,” 1989.